



United International University
School of Science and Engineering
Final Exam Trimester: Spring 2025
Course Title: Coordinate Geometry and Vector Analysis
Course Code: Math 2201 Marks: 40
Total Time: 2 hours

Answer all questions.

1.	a) Use the Divergence Theorem to find the outward flux of the vector field $F(x, y, z) = 2x^3\mathbf{i} + 2y^3\mathbf{j} + 2z^3\mathbf{k}$ across the surface of the region that is enclosed by $z = \sqrt{1 - x^2 - y^2}$ and the plane $z = 0$. [5] b) Evaluate the surface integral $\iint_{\sigma} 25x^2z^2 \, ds$; σ is the part of the cone $z = \sqrt{x^2 + y^2}$ that lies between the planes $z = 1$ and $z = 3$. [5]
✓ 2.	a) Use cylindrical coordinate systems to evaluate: $\int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_0^{16-2(x^2+y^2)} (x^2 + y^2) \, dz \, dy \, dx$ [5] b) Identify and Sketch the Conic $9x^2 + 4y^2 - 18x + 24y = -9$ [5]
✓ 3.	a) Consider, $F(x, y) = (3x^2 + 2xy^2)\mathbf{i} + (2x^2y + 3y^2)\mathbf{j}$ [5] i) Show that F is a conservative vector field on the entire xy-plane. ii) Find the potential function $\phi(x, y)$. iii) Find $\int_{(0,0)}^{(1,1)} F \cdot d\mathbf{r}$ using ii) b) Using Green's theorem find the value of $\oint_C F \cdot d\mathbf{r}$ [5] Where $F(x, y) = (e^x - 2y^2)\mathbf{i} + (\cos y + x^2)\mathbf{j}$ and C is the closed triangle bounded by the lines $x = 0$, $y = 0$ and $x + y = 4$.
4.	a) Evaluate $\int_C (2x - y)dx + (y - x)dy$ along the triangle with vertices $(0, 0)$, $(0, 2)$ and $(2, 0)$. [5] b) Use triple integral to find the volume of the solid G enclosed by the sphere $x^2 + y^2 + z^2 = 4$ and the cone $z = \sqrt{x^2 + y^2}$. [5]